



UNIVERSITY OF
WATERLOO

MTE 203 – Advanced Calculus

Midterm Exam Fall 2016 – Problem 3

Midterm Exam Fall 2016 – Problem 3

If $z = f(x, y)$ has continuous second-order partial derivatives and

$$x = r^2 + s^2$$

$$y = 2rs$$

Use the chain rule to find,

$$\frac{\partial^2 z}{\partial r^2}$$

Solution

Index: Chain Rule



UNIVERSITY OF
WATERLOO

Midterm Exam Fall 2016 – Problem 3

Midterm Exam Fall 2016 – Problem 3

Midterm Exam Fall 2016 – Problem 3

Midterm Exam Fall 2016 – Problem 3

Midterm Exam Fall 2016 – Problem 3